NumPy Searching Arrays

Searching Arrays

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5, 4, 4])

x = np.where(arr == 4)

print(x)
```

```
(array([3, 5, 6]),)
```

Find the indexes where the values are odd:

```
import numpy as np

arr = np.array([10, 14, 93, 41, 8, 7])

x = np.where(arr%2 == 1)

print(x)
```

```
(array([2, 3, 5]),)
```

Find the indexes where the values are even:

```
import numpy as np

arr = np.array([10, 14, 93, 41, 8, 7])

x = np.where(arr%2 == 0)

print(x)
```

```
(array([0, 1, 4]),)
```

Search Sorted

```
import numpy as np

arr = np.array([6, 7, 8, 9])

x = np.searchsorted(arr, 7)

print(x)
```

```
1
```

Search From the Right Side

```
import numpy as np

arr = np.array([6, 7, 8, 9])

x = np.searchsorted(arr, 7, side='right')

print(x)
```

```
2
```

Multiple Values

```
import numpy as np

arr = np.array([1, 3, 5, 7])

x = np.searchsorted(arr, [2, 4, 6])

print(x)
```

```
[1 2 3]
```